Eco-Friendly Alternatives to Disposable Plastic Foodware in Sri Lanka

A Comparative Analysis of Consumer Perspectives, Cost, Durability, Convenience and Life-Cycle Performance

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Abstract

The use of disposable foodware creates environmental challenges, while consumers and food-service providers find it difficult to resist its affordability and convenience. This study proposes a quantitative investigation of consumer perception of alternatives to conventional disposable plastic foodware in Sri Lanka. The study focuses on food-service items such as spoons, plates, and straws, and assesses alternatives including plant-based products, bioplastics, stainless steel and glass. Consumer perceptions will be assessed through a structured questionnaire covering usage patterns, environmental awareness, perceived cost, durability, convenience, hygiene, availability, environmental impact and overall satisfaction.

To validate the consumer analysis, the study incorporates a secondary-data benchmark developed in the accompanying research dataset. This benchmark compares standardized food-service items using Sri Lankan market prices, cost per use, expected lifespan, production-footprint scores, end-of-life scores, durability scores and lifespan scores. A provisional weighted scoring framework assigns 15% to cost, 30% to production footprint, 25% to end-of-life performance, 10% to durability and 20% to lifespan. This benchmark indicates that stainless steel receives the highest score among the listed materials, while conventional plastic receives the lowest.

The final study will integrate the consumer survey with product-level evidence to determine which alternatives are most acceptable and practically suitable for different foodware applications in Sri Lanka.

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1. Introduction

1.1 Background

Disposable foodware is widely used as it is inexpensive and convenient. However, when products are discarded after a short period of use, it contributes to a difficult waste-management problem. Reviews of single-use plastics identify widespread use in packaging and disposable tableware and describe pollution, resource loss and limitations in recycling as major concerns [1,2].

The sustainability of an alternative cannot be judged from material alone. The United Nations Environment Programme's life-cycle review stresses that the number of times a product is used is a critical consideration and identifies reusable systems as important options in reducing single-use product impacts [11]. Likewise, research on packaging shows that consumers may use visible characteristics as cues for sustainability, while their perceptions do not necessarily correspond to full life-cycle evidence [9].

For foodware, alternatives need to be considered through both environmental and consumer perspectives. Plant-fibre and bamboo products may offer biodegradable characteristics in particular applications; bioplastics may replicate some of the functionality of conventional plastics but have material and disposal-specific limitations, and reusable stainless steel and glass can have long service lives but require repeated cleaning and transport.

1.2 Problem Statement

Although sustainable foodware alternatives are increasingly available to consumers, there is insufficient consumer-centred evidence for determining which materials are considered acceptable in the Sri Lankan context. Existing literature frequently examines individual materials or technical and environmental performance, whereas consumers make decisions using several criteria simultaneously.

The problem addressed by this study is therefore the absence of an integrated Sri Lankan comparison that connects consumer preferences with cost, durability, lifespan and life-cycle-oriented evidence across different foodware categories.

1.3 Aim

To comparatively evaluate consumer perceptions and environmental performance indicators of eco-friendly alternatives to disposable plastic foodware in Sri Lanka.

1.4 Research Objectives

- To determine patterns of consumer use of single-use foodware in Sri Lanka.
- To assess consumer awareness of the environmental impact of disposable plastic foodware.
- To compare consumer perceptions of plant-based products, bioplastics, stainless steel and glass.
- To evaluate the importance of cost, durability, convenience, hygiene, availability, appearance and perceived environmental impact in consumer choice.
- To compare selected foodware alternatives using Sri Lankan market price, cost per use, and life cycle oriented indicators.
- To identify the alternatives that appear most suitable for different foodware applications.
- To develop evidence-based recommendations for consumers, food-service businesses and future research.

1.5 Research Questions

- How frequently do consumers in Sri Lanka use single-use foodware?
- How aware are consumers of the environmental impacts of disposable plastic foodware?
- Which eco-friendly foodware materials are most preferred by consumers?
- How do consumers rank cost, durability, convenience, hygiene, availability and environmental impact when selecting alternatives?
- How do the consumer preferences compare with cost per use and life-cycle-oriented performance indicators?
- Which alternatives appear most suitable for different foodware categories?

1.6 Research Hypotheses

- Consumer preference for eco-friendly foodware is positively associated with environmental awareness.
- Perceived affordability and convenience are positively associated with willingness to adopt eco-friendly foodware.
- Perceived durability is positively associated with overall satisfaction with reusable foodware alternatives.
- Consumers' perceived environmental performance may not correspond directly to the life-cycle-oriented ranking of alternatives.

1.7 Scope of the Study

The study is limited to foodware used for eating and drinking. The primary categories are cutlery, crockery, and straws. The primary component covers respondents in Sri Lanka using a structured questionnaire. The secondary component uses Sri Lankan market-price observations and international secondary environmental evidence. The study does not claim to perform a new ISO-certified life-cycle assessment, laboratory durability test or controlled environmental experiment.

1.8 Significance of the Study

This research can help provide a broader view of sustainable foodware choices. A product should not necessarily be considered the best option simply because it is described as biodegradable, plant-based or reusable. Other factors, including how long it can be used, its cost, convenience and disposal requirements, also need to be considered.

For consumers, the study may help show the differences between the available alternatives and the trade-offs involved in choosing them. For food-service businesses, the findings may be useful when selecting materials for different applications. The research may also be useful to future researchers who want to compare consumer opinions with cost-per-use and life-cycle-related information in Sri Lanka.

2. Literature Review

2.1 Environmental Impacts of Single-Use Plastic Foodware

Single-use plastics are designed for short term use and are widely used in packaging and disposable tableware. Their low cost and functional performance have encouraged widespread adoption, while disposal and inadequate recovery can lead to persistent waste and environmental pollution [1,2]. Food packaging is a particularly important area because plastics provide useful barrier and protection functions while generating waste after their intended use [3].

2.2 Paper and Wooden Alternatives

Paper and wooden products can reduce reliance on conventional plastic in selected applications, but environmental desirability does not guarantee equivalent functional performance. Research on paper straws has documented substantial reductions in compressive strength after liquid exposure, illustrating how moisture resistance and use duration can influence consumer acceptance [4].

2.3 Plant-Based Foodware

Research on areca palm sheath has examined its mechanical and diffusion properties and its potential for foodware applications [5]. However, the environmental outcome depends on production, transport, use and end-of-life conditions rather than biodegradability alone.

2.4 Bioplastics

Bioplastics are not a single material class. Bio-based origin and biodegradability are separate properties, and a bioplastic's environmental performance depends on its composition, production pathway and end-of-life conditions [6]. Recent review evidence also emphasizes that many biodegradable plastics require specific conditions for effective degradation and that inadequate sorting, collection and treatment infrastructure can reduce their environmental advantages [6].

2.5 Stainless Steel and Glass

Stainless steel and glass are reusable materials with potentially long service lives. Their sustainability therefore depends strongly on how many times they are actually reused and on the environmental burdens associated with production, cleaning and transport. Stainless steel can provide high durability and repeated-use potential, while glass is valued for its inertness and reusability but can be heavier and more fragile in some applications. The dataset reflects these trade-offs by assigning higher provisional durability and lifespan scores to the reusable products.

2.6 Consumer Acceptance and the Attitude–Behaviour Gap

Consumer acceptance is shaped by more than environmental concern. Packaging research indicates that sustainability can be salient to consumers without being the strongest determinant of product attitudes, and consumers may rely on material cues that do not accurately represent life-cycle performance [9].

2.7 Recent Policy Changes

National policy interventions in Sri Lanka, such as bans on single-use polythene and mandatory pricing on plastic carrier bags, have demonstrated that economic disincentives directly alter consumer habits. Expanding these economic instruments to food-service disposables requires empirical benchmarks to ensure suitable, affordable alternatives exist in the local market. [11].

2.8 Research Gap

The literature establishes the environmental concerns associated with single-use plastics and provides evidence on the technical and life-cycle characteristics of several alternatives. However, the present research addresses a more specific gap, a Sri Lankan consumer comparison across several foodware categories, linked to a local-market benchmark. The study therefore combines consumer perceptions with cost per use, lifespan and life cycle oriented indicators instead of treating environmental performance as the sole criterion.

3. Conceptual Framework

The proposed framework assumes that adoption of eco-friendly foodware is influenced by both individual perceptions and observable product characteristics. Environmental awareness may increase willingness to adopt, but cost, convenience, durability, hygiene and availability can strengthen or weaken that intention.

Input / Independent factors	Mediating perceptions	Outcome
Environmental awareness	Perceived environmental benefit	Preference for alternative
Price / affordability	Perceived value	Willingness to adopt
Durability / lifespan	Perceived performance	Overall satisfaction
Convenience / weight / cleaning	Perceived usability	Intention to reuse
Availability / accessibility	Perceived feasibility	Purchase intention

The secondary-data component complements this framework by providing product-level indicators for cost per use, production footprint, end-of-life performance, durability and lifespan. The two evidence streams are compared, not mathematically treated as interchangeable.

4. Methodology

4.1 Research Design

A quantitative, cross-sectional research design will be used. The primary component is a structured consumer survey. The secondary component is a comparative product benchmark using Sri Lankan market prices and selected secondary life cycle evidence. This design allows the research to compare what consumers prefer with what product-level evidence suggests about cost and repeated use.

4.2 Population and Sampling

The target population comprises consumers in Sri Lanka who use food-service products such as plates, cutlery and straws. The existing draft reports approximately 50 respondents, mainly university students, with respondents from other age groups and backgrounds. Convenience sampling was used for the preliminary survey. Because this is a non-probability sample, the findings should not be generalized to the entire Sri Lankan population without qualification.

4.3 Data Collection Instrument

The questionnaire was designed in Google Forms and uses multiple-choice, checkbox and rating-scale questions. The instrument covers demographic characteristics, frequency of single-use foodware use, commonly used items, environmental awareness, preferences, willingness to adopt alternatives, and evaluations of different materials. Evaluation criteria include cost, durability, appearance, environmental impact, hygiene, availability, convenience and overall satisfaction.

4.4 Secondary Product Benchmark

The accompanying research dataset provides 15 standardized observations covering spoons, dinner plates and drinking straws. It records LKR per unit, expected uses, LKR per use, production-footprint score, end-of-life score, durability score, lifespan score, cost score and a provisional weighted score. The proposed functional unit is one use of one standardized food-service item. The dataset specifies approximate product dimensions.

4.5 Cost-per-Use Calculation

For reusable items, cost per use is calculated as LKR per unit divided by expected number of uses. Single-use products are assigned one use.

4.6 Preliminary Weighted Scoring Model

The accompanying dataset specifies the following provisional weights: cost = 15%, production footprint = 30%, end-of-life = 25%, durability = 10% and lifespan = 20%. Each component is scored from 1 to 5. The weighted score is calculated as;

Weighted Score = 0.15 (Cost) + 0.30 (Production) + 0.25 (EOL) + 0.10 (Durability) + 0.20 (Lifespan)

Higher scores indicate better performance within the provisional scoring framework.

5. Secondary Data and Analysis

5.1 Product-Level Benchmark

The table below reproduces the core product-level data from the additional dataset in a condensed form.

Product	Material	LKR/unit	LKR/use*	Expected uses	Weighted score /5
Spoon	Glass	746.67	7.46	100	2.9
Spoon	Stainless steel	120.00	0.30	400	4.1
Spoon	Bamboo	113.00	1.50	75	3.7
Spoon	Bio-plastic (PLA/CPLA)	63.02	6.30	10	2.4
Spoon	Conventional plastic	29.17	2.91	10	1.6
Dinner plate	Glass	741.67	1.48	500	3.6
Dinner plate	Stainless steel	499.50	0.33	1500	4.1
Dinner plate	Bamboo	220.20	0.88	250	3.7
Dinner plate	Bio-plastic (PLA/CPLA)	88.00	3.52	25	2.4
Dinner plate	Conventional plastic	250.00	5.00	50	1.6
Drinking straw	Glass	268.00	2.68	100	3.6
Drinking straw	Stainless steel	140.00	0.28	500	4.1
Drinking straw	Bamboo	128.00	6.40	50	3.4
Drinking straw	Bio-plastic (PLA)	56.72	5.67	10	2.7
Drinking straw	Conventional plastic	10.26	10.26	1	1.3

*Cost per use before any additional lifespan adjustment.

5.2 Material-Level Summary

Material	Avg. LKR/unit*	Avg. LKR/use*	Avg. provisional score	Interpretation
Stainless steel	253.17	0.30	4.10	Strongest overall result; high expected reuse
Bamboo	153.73	2.93	3.60	Intermediate result; plant-based option with category-specific trade-offs
Glass	585.45	3.87	3.37	Reusable but higher unit cost and category-dependent practicality
Bioplastic (PLA/PLA-CPLA)	71	3.8	2.55	Performance varies by product and disposal pathway
Conventional plastic	96.48	6.06	1.50	Lowest score in supplied benchmark

*Simple averages across the supplied observations; they are descriptive only and should not be treated as population-weighted estimates.

5.3 Interpretation of Results

The preliminary benchmark shows a clear difference between the materials included in the dataset. Stainless steel has the highest provisional weighted score among the materials represented in all three product categories, while conventional plastic has the lowest score. Stainless steel also has a low calculated cost per use because the dataset assumes that the products can be used many more times than the disposable alternatives.

Bamboo falls between the reusable metal products and the lower-scoring alternatives. Glass is also reusable, but its higher purchase price results in a higher cost per use in the supplied data.

These results should be treated as preliminary rather than as proof that one material is always environmentally better than another. The ranking is affected by the weights used in the model, the expected number of uses, the provisional scores and the particular products included in the dataset.

5.4 Relationship to the Consumer Survey

The survey reports that approximately 50 respondents participated, mainly university students, and that bioplastics were the most preferred option in the survey, with plant-based products receiving positive responses for their natural and biodegradable characteristics. Stainless steel and glass were valued for durability and reusability but raised concerns about cost and convenience. These reported survey findings differ from the product benchmark, which gives stainless steel the highest provisional score. This gap highlights a clear misalignment between what consumers prefer and what the lifecycle data shows.

6. Expected Findings

Based on the survey findings and the product benchmark, the study is expected to indicate that consumers evaluate sustainable foodware through a combination of environmental and practical criteria. It suggests relatively strong consumer interest in bioplastics and positive perceptions towards plant-based products, while reusable stainless steel and glass are associated with durability and reusability but also with cost and convenience concerns.

The secondary benchmark indicates that stainless steel receives the highest provisional weighted score in the supplied dataset, largely because of its assumed lifespan and low cost per use. This indicates that the most preferred material from a consumer perspective may not be the highest-scoring option under a life-cycle-oriented decision framework.

7. Limitations

- The preliminary consumer sample is small and convenience-based.
- Expected lifespan values for reusable products are partly assumed.
- Consumer self-reported preferences may differ from actual purchasing behaviour.
- The study does not experimentally test durability, hygiene or degradation.

8. Conclusion

To truly move away from single-use plastics in Sri Lanka, we cannot just rely on what looks "green" on paper or what feels convenient at a food counter. While the survey indicates that people naturally lean toward bioplastics and single-use plant items because it fits right into their daily takeaway habits, the pricing and lifecycle data proves that reusables, especially stainless steel, are by far the most cost-effective and sustainable option over time. The real challenge is not just offering alternative materials, it is figuring out how to build the washing infrastructure, vendor setups, and local waste systems needed to make those alternatives actually work in everyday Sri Lankan food service. Moving forward, the team will focus on connecting these survey insights with real retail pricing to offer practical, category-specific guidelines for local vendors.

References

[1] Chen, Y., Awasthi, A. K., Wei, F., Tan, Q., & Li, J. (2021). Single-use plastics: Production, usage, disposal, and adverse impacts. Science of the Total Environment, 752, 141772. https://doi.org/10.1016/j.scitotenv.2020.141772

[2] Dey, A., Dhumal, C. V., Sengupta, P., Kumar, A., Pramanik, N. K., & Alam, T. (2021). Challenges and possible solutions to mitigate the problems of single-use plastics used for packaging food items: A review. Journal of Food Science and Technology, 58, 3251–3269. https://doi.org/10.1007/s13197-020-04885-6

[3] Ncube, L. K., Ude, A. U., Ogunmuyiwa, E. N., Zulkifli, R., & Beas, I. N. (2020). Environmental impact of food packaging materials: A review of contemporary development from conventional plastics to polylactic acid based materials. Materials, 13(21), 4994. https://doi.org/10.3390/ma13214994

[4] Gutierrez, J. N., Royals, A. W., Jameel, H., Venditti, R. A., & Pal, L. (2019). Evaluation of paper straws versus plastic straws: Development of a methodology for testing and understanding challenges for paper straws. BioResources, 14(4), 8345–8363. https://doi.org/10.15376/biores.14.4.8345-8363

[5] Mohanty, D. P. (2021). Areca palm sheath: A plant-based material alternative to plastics for foodware products [Doctoral dissertation, Purdue University]. Purdue e-Pubs. https://doi.org/10.25394/PGS.17129708.v1

[6] Hwang, E., Yang, Y.-H., Choi, J., Park, S.-H., Park, K., & Lee, J. (2025). Biodegradable plastics as sustainable alternatives: Advances, basics, challenges, and directions for the future. Materials, 18(18), 4247. https://doi.org/10.3390/ma18184247

[7] United Nations Environment Programme. (2021). Addressing single-use plastic products pollution using a life cycle approach. UNEP.

[8] Research dataset supplied for this study. (2026). Sri Lanka single-use food products research dataset [Unpublished preliminary research database].

[9] Steenis, N. D., van Herpen, E., van der Lans, I. A., Ligthart, T. N., & van Trijp, H. C. M. (2017). Consumer response to packaging design: The role of packaging materials and graphics in sustainability perceptions and product evaluations. Journal of Cleaner Production, 162, 286–298. https://doi.org/10.1016/j.jclepro.2017.06.036

[10] Life Cycle Initiative. (2021). Addressing single-use plastic products pollution using a life cycle approach. United Nations Environment Programme / Life Cycle Initiative.

[11] "CAA Announces New Rules on Polyethene Bags and Bottles," News.lk, Oct. 2, 2025. [Online]. Available: https://news.lk/current-affairs/caa-announces-new-rules-on-polyethene-bags-and-bottles. [Accessed: Aug. 7, 2026].